

# Track 2 Multi-Model Curve Comparison Report

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

## Overview

This report compares representative Track 2 TE curves by overlaying multiple model predictions on the same original measured curve. The plots are intended to show whether each model tracks the local harmonic oscillations rather than only the broad mean trend.

## Scope

- each comparison image contains four deterministic held-out test curves;
- forward comparisons are shown on forward curves only;
- backward comparisons are shown on backward curves only;
- Wave 1 screening keeps the three strongest family-best models by Curve MAE [deg] within each direction;
- Original Curve uses the same visual weight as predictions and a dark-gray color for balanced comparison.

## Metrics Summary

### Forward Reference Model Overlay

| Candidate              | Source        | Surface | Curve MAE [deg] | Curve RMSE [deg] | Mean Error |
|------------------------|---------------|---------|-----------------|------------------|------------|
| paper_original_best_Fw | rcim_original | Fw      | 0.002769        | 0.002951         | 6.250      |
| paper_retuned_best_Fw  | rcim_retuned  | Fw      | 0.001839        | 0.002041         | 4.109      |
| track1_best_Fw         | rcim_track1   | Fw      | 0.003014        | 0.003204         | 6.819      |

### Forward Wave 1 Family Model Overlay

| Candidate                | Source                               | Surface | Curve MAE [deg] | Curve RMSE [deg] | Mean Error |
|--------------------------|--------------------------------------|---------|-----------------|------------------|------------|
| feedforward_fw           | wave1_current_registry               | Fw      | 0.003404        | 0.003855         | 7.551      |
| harmonic_regression_fw   | wave1_current_registry               | Fw      | 0.003230        | 0.003494         | 7.185      |
| periodic_mlp_fw          | wave1_current_registry               | Fw      | 0.003254        | 0.003553         | 7.232      |
| residual_harmonic_mlp_fw | wave1_current_registry               | Fw      | 0.003273        | 0.003563         | 7.266      |
| tree_fw                  | wave1_current_registry               | Fw      | 0.003053        | 0.003395         | 6.731      |
| periodic_mlp_harmonic_fw | wave1_periodic_mlp_harmonic_campaign | Fw      | 0.003254        | 0.003553         | 7.232      |

Backward Reference Model Overlay

| Candidate             | Source       | Surface | Curve MAE [deg] | Curve RMSE [deg] | Mean Error |
|-----------------------|--------------|---------|-----------------|------------------|------------|
| paper_retuned_best_Bw | rcim_retuned | Bw      | 0.003675        | 0.004284         | 7.572      |
| track1_best_Bw        | rcim_track1  | Bw      | 0.005027        | 0.005212         | 11.860     |

Backward Wave 1 Family Model Overlay

| Candidate                | Source                               | Surface | Curve MAE [deg] | Curve RMSE [deg] | Mean Error |
|--------------------------|--------------------------------------|---------|-----------------|------------------|------------|
| feedforward_bw           | wave1_current_registry               | Bw      | 0.003586        | 0.004023         | 7.832      |
| harmonic_regression_bw   | wave1_current_registry               | Bw      | 0.003678        | 0.004012         | 8.058      |
| periodic_mlp_bw          | wave1_current_registry               | Bw      | 0.003574        | 0.004006         | 7.807      |
| residual_harmonic_mlp_bw | wave1_current_registry               | Bw      | 0.003536        | 0.003874         | 7.728      |
| tree_bw                  | wave1_current_registry               | Bw      | 0.003258        | 0.003651         | 7.051      |
| periodic_mlp_harmonic_bw | wave1_periodic_mlp_harmonic_campaign | Bw      | 0.003583        | 0.003925         | 7.875      |

Forward Wave 2 Temporal Model Overlay

| Candidate                        | Source                        | Surface | Curve MAE [deg] | Curve RMSE [deg] | Mean Error |
|----------------------------------|-------------------------------|---------|-----------------|------------------|------------|
| temporal_convolution_fw          | wave2_temporal_entry_registry | Fw      | 0.003603        | 0.004031         | 8.028      |
| gru_sequence_fw                  | wave2_temporal_entry_registry | Fw      | 0.003330        | 0.003762         | 7.378      |
| lstm_sequence_fw                 | wave2_temporal_entry_registry | Fw      | 0.003366        | 0.003800         | 7.450      |
| periodic_temporal_convolution_fw | wave2_temporal_entry_registry | Fw      | 0.003335        | 0.003708         | 7.404      |
| periodic_gru_sequence_fw         | wave2_temporal_entry_registry | Fw      | 0.003186        | 0.003438         | 7.077      |
| periodic_lstm_sequence_fw        | wave2_temporal_entry_registry | Fw      | 0.003266        | 0.003550         | 7.258      |

Backward Wave 2 Temporal Model Overlay

| Candidate                        | Source                        | Surface | Curve MAE [deg] | Curve RMSE [deg] | Mean Error |
|----------------------------------|-------------------------------|---------|-----------------|------------------|------------|
| temporal_convolution_bw          | wave2_temporal_entry_registry | Bw      | 0.003742        | 0.004166         | 8.184      |
| gru_sequence_bw                  | wave2_temporal_entry_registry | Bw      | 0.003626        | 0.004082         | 7.907      |
| lstm_sequence_bw                 | wave2_temporal_entry_registry | Bw      | 0.003555        | 0.003985         | 7.767      |
| periodic_temporal_convolution_bw | wave2_temporal_entry_registry | Bw      | 0.003628        | 0.003987         | 7.979      |
| periodic_gru_sequence_bw         | wave2_temporal_entry_registry | Bw      | 0.002392        | 0.002639         | 5.466      |
| periodic_lstm_sequence_bw        | wave2_temporal_entry_registry | Bw      | 0.002625        | 0.002877         | 6.013      |

### Forward Wave 2C Residual Harmonic Temporal Overlay

| Candidate                                      | Source                                     | Surface | Curve MAE [deg] | Curve RMSE [deg] | Mean Error |
|------------------------------------------------|--------------------------------------------|---------|-----------------|------------------|------------|
| residual_harmonic_gru_sequence_sparse_rcim_Fw  | wave2c_residual_harmonic_temporal_registry | Fw      | 0.003194        | 0.003499         | 7.083      |
| residual_harmonic_gru_sequence_dense240_Fw     | wave2c_residual_harmonic_temporal_registry | Fw      | 0.006983        | 0.009275         | 15.722     |
| residual_harmonic_gru_sequence_dense360_Fw     | wave2c_residual_harmonic_temporal_registry | Fw      | 0.007869        | 0.010574         | 17.740     |
| residual_harmonic_lstm_sequence_sparse_rcim_Fw | wave2c_residual_harmonic_temporal_registry | Fw      | 0.003229        | 0.003533         | 7.164      |
| residual_harmonic_lstm_sequence_dense240_Fw    | wave2c_residual_harmonic_temporal_registry | Fw      | 0.007042        | 0.009370         | 15.868     |
| residual_harmonic_lstm_sequence_dense360_Fw    | wave2c_residual_harmonic_temporal_registry | Fw      | 0.007731        | 0.010235         | 17.430     |

### Backward Wave 2C Residual Harmonic Temporal Overlay

| Candidate                                      | Source                                     | Surface | Curve MAE [deg] | Curve RMSE [deg] | Mean Error |
|------------------------------------------------|--------------------------------------------|---------|-----------------|------------------|------------|
| residual_harmonic_gru_sequence_sparse_rcim_Bw  | wave2c_residual_harmonic_temporal_registry | Bw      | 0.003502        | 0.003857         | 7.654      |
| residual_harmonic_gru_sequence_dense240_Bw     | wave2c_residual_harmonic_temporal_registry | Bw      | 0.008984        | 0.012987         | 20.358     |
| residual_harmonic_gru_sequence_dense360_Bw     | wave2c_residual_harmonic_temporal_registry | Bw      | 0.009370        | 0.013165         | 21.267     |
| residual_harmonic_lstm_sequence_sparse_rcim_Bw | wave2c_residual_harmonic_temporal_registry | Bw      | 0.003440        | 0.003793         | 7.510      |
| residual_harmonic_lstm_sequence_dense240_Bw    | wave2c_residual_harmonic_temporal_registry | Bw      | 0.007367        | 0.009945         | 16.660     |
| residual_harmonic_lstm_sequence_dense360_Bw    | wave2c_residual_harmonic_temporal_registry | Bw      | 0.010268        | 0.014769         | 23.355     |

### Forward Track 1 And Screened Wave 1 Overlay

| Candidate              | Source                 | Surface | Curve MAE [deg] | Curve RMSE [deg] | Mean Error |
|------------------------|------------------------|---------|-----------------|------------------|------------|
| track1_best_Fw         | rcim_track1            | Fw      | 0.003014        | 0.003204         | 6.819      |
| tree_fw                | wave1_current_registry | Fw      | 0.003053        | 0.003395         | 6.731      |
| harmonic_regression_fw | wave1_current_registry | Fw      | 0.003230        | 0.003494         | 7.185      |
| periodic_mlp_fw        | wave1_current_registry | Fw      | 0.003254        | 0.003553         | 7.232      |

### Backward Track 1 And Screened Wave 1 Overlay

| Candidate                | Source                 | Surface | Curve MAE [deg] | Curve RMSE [deg] | Mean Error |
|--------------------------|------------------------|---------|-----------------|------------------|------------|
| track1_best_Bw           | rcim_track1            | Bw      | 0.005027        | 0.005212         | 11.860     |
| tree_bw                  | wave1_current_registry | Bw      | 0.003258        | 0.003651         | 7.051      |
| residual_harmonic_mlp_bw | wave1_current_registry | Bw      | 0.003536        | 0.003874         | 7.728      |
| periodic_mlp_bw          | wave1_current_registry | Bw      | 0.003574        | 0.004006         | 7.807      |

### Forward Reference Tree And Wave 2 Overlay

| Candidate                        | Source                        | Surface | Curve MAE [deg] | Curve RMSE [deg] | Mean Error |
|----------------------------------|-------------------------------|---------|-----------------|------------------|------------|
| paper_retuned_best_Fw            | rcim_retuned                  | Fw      | 0.001839        | 0.002041         | 4.109      |
| track1_best_Fw                   | rcim_track1                   | Fw      | 0.003014        | 0.003204         | 6.819      |
| tree_fw                          | wave1_current_registry        | Fw      | 0.003053        | 0.003395         | 6.731      |
| temporal_convolution_fw          | wave2_temporal_entry_registry | Fw      | 0.003603        | 0.004031         | 8.028      |
| gru_sequence_fw                  | wave2_temporal_entry_registry | Fw      | 0.003330        | 0.003762         | 7.378      |
| lstm_sequence_fw                 | wave2_temporal_entry_registry | Fw      | 0.003366        | 0.003800         | 7.450      |
| periodic_temporal_convolution_fw | wave2_temporal_entry_registry | Fw      | 0.003335        | 0.003708         | 7.404      |
| periodic_gru_sequence_fw         | wave2_temporal_entry_registry | Fw      | 0.003186        | 0.003438         | 7.077      |
| periodic_lstm_sequence_fw        | wave2_temporal_entry_registry | Fw      | 0.003266        | 0.003550         | 7.258      |

### Backward Reference Tree And Wave 2 Overlay

| Candidate                        | Source                        | Surface | Curve MAE [deg] | Curve RMSE [deg] | Mean Error |
|----------------------------------|-------------------------------|---------|-----------------|------------------|------------|
| paper_retuned_best_Bw            | rcim_retuned                  | Bw      | 0.003675        | 0.004284         | 7.572      |
| track1_best_Bw                   | rcim_track1                   | Bw      | 0.005027        | 0.005212         | 11.860     |
| tree_bw                          | wave1_current_registry        | Bw      | 0.003258        | 0.003651         | 7.051      |
| temporal_convolution_bw          | wave2_temporal_entry_registry | Bw      | 0.003742        | 0.004166         | 8.184      |
| gru_sequence_bw                  | wave2_temporal_entry_registry | Bw      | 0.003626        | 0.004082         | 7.907      |
| lstm_sequence_bw                 | wave2_temporal_entry_registry | Bw      | 0.003555        | 0.003985         | 7.767      |
| periodic_temporal_convolution_bw | wave2_temporal_entry_registry | Bw      | 0.003628        | 0.003987         | 7.979      |
| periodic_gru_sequence_bw         | wave2_temporal_entry_registry | Bw      | 0.002392        | 0.002639         | 5.466      |
| periodic_lstm_sequence_bw        | wave2_temporal_entry_registry | Bw      | 0.002625        | 0.002877         | 6.013      |

### Forward Reference Tree And Wave 2C Overlay

| Candidate                                      | Source                                     | Surface | Curve MAE [deg] | Curve RMSE [deg] | Mean Error |
|------------------------------------------------|--------------------------------------------|---------|-----------------|------------------|------------|
| paper_retuned_best_Fw                          | rcim_retuned                               | Fw      | 0.001839        | 0.002041         | 4.109      |
| track1_best_Fw                                 | rcim_track1                                | Fw      | 0.003014        | 0.003204         | 6.819      |
| tree_fw                                        | wave1_current_registry                     | Fw      | 0.003053        | 0.003395         | 6.731      |
| residual_harmonic_gru_sequence_sparse_rcim_Fw  | wave2c_residual_harmonic_temporal_registry | Fw      | 0.003194        | 0.003499         | 7.083      |
| residual_harmonic_gru_sequence_dense240_Fw     | wave2c_residual_harmonic_temporal_registry | Fw      | 0.006983        | 0.009275         | 15.722     |
| residual_harmonic_gru_sequence_dense360_Fw     | wave2c_residual_harmonic_temporal_registry | Fw      | 0.007869        | 0.010574         | 17.740     |
| residual_harmonic_lstm_sequence_sparse_rcim_Fw | wave2c_residual_harmonic_temporal_registry | Fw      | 0.003229        | 0.003533         | 7.164      |
| residual_harmonic_lstm_sequence_dense240_Fw    | wave2c_residual_harmonic_temporal_registry | Fw      | 0.007042        | 0.009370         | 15.868     |
| residual_harmonic_lstm_sequence_dense360_Fw    | wave2c_residual_harmonic_temporal_registry | Fw      | 0.007731        | 0.010235         | 17.430     |



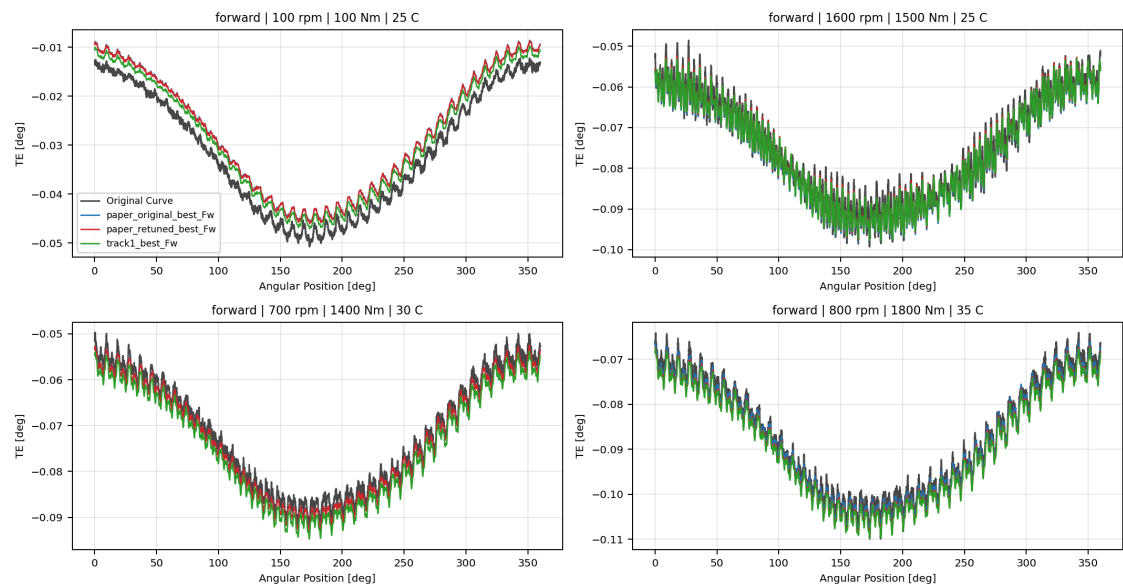
Backward Reference Tree And Wave 2C Overlay

| Candidate                                      | Source                                     | Surface | Curve MAE [deg] | Curve RMSE [deg] | Mean Error |
|------------------------------------------------|--------------------------------------------|---------|-----------------|------------------|------------|
| paper_retuned_best_Bw                          | rcim_retuned                               | Bw      | 0.003675        | 0.004284         | 7.572      |
| track1_best_Bw                                 | rcim_track1                                | Bw      | 0.005027        | 0.005212         | 11.860     |
| tree_bw                                        | wave1_current_registry                     | Bw      | 0.003258        | 0.003651         | 7.051      |
| residual_harmonic_gru_sequence_sparse_rcim_Bw  | wave2c_residual_harmonic_temporal_registry | Bw      | 0.003502        | 0.003857         | 7.654      |
| residual_harmonic_gru_sequence_dense240_Bw     | wave2c_residual_harmonic_temporal_registry | Bw      | 0.008984        | 0.012987         | 20.358     |
| residual_harmonic_gru_sequence_dense360_Bw     | wave2c_residual_harmonic_temporal_registry | Bw      | 0.009370        | 0.013165         | 21.267     |
| residual_harmonic_lstm_sequence_sparse_rcim_Bw | wave2c_residual_harmonic_temporal_registry | Bw      | 0.003440        | 0.003793         | 7.510      |
| residual_harmonic_lstm_sequence_dense240_Bw    | wave2c_residual_harmonic_temporal_registry | Bw      | 0.007367        | 0.009945         | 16.660     |
| residual_harmonic_lstm_sequence_dense360_Bw    | wave2c_residual_harmonic_temporal_registry | Bw      | 0.010268        | 0.014769         | 23.355     |

# Comparison Gallery - Forward Reference Model Overlay

Included models: `paper_original_best_Fw` , `paper_retuned_best_Fw` , `track1_best_Fw` .

Forward Reference Model Overlay

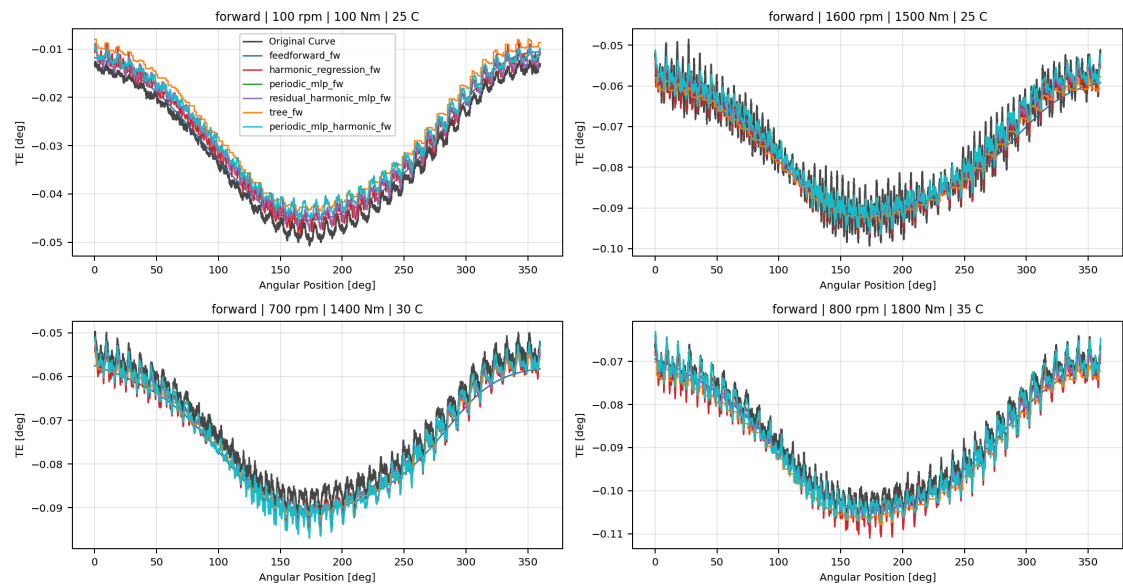


Forward Reference Model Overlay Track 2 comparison

# Comparison Gallery - Forward Wave 1 Family Model Overlay

Included models: `feedforward_fw` , `harmonic_regression_fw` , `periodic_mlp_fw` , `residual_harmonic_mlp_fw` , `tree_fw` , `periodic_mlp_harmonic_fw` .

Forward Wave 1 Family Model Overlay

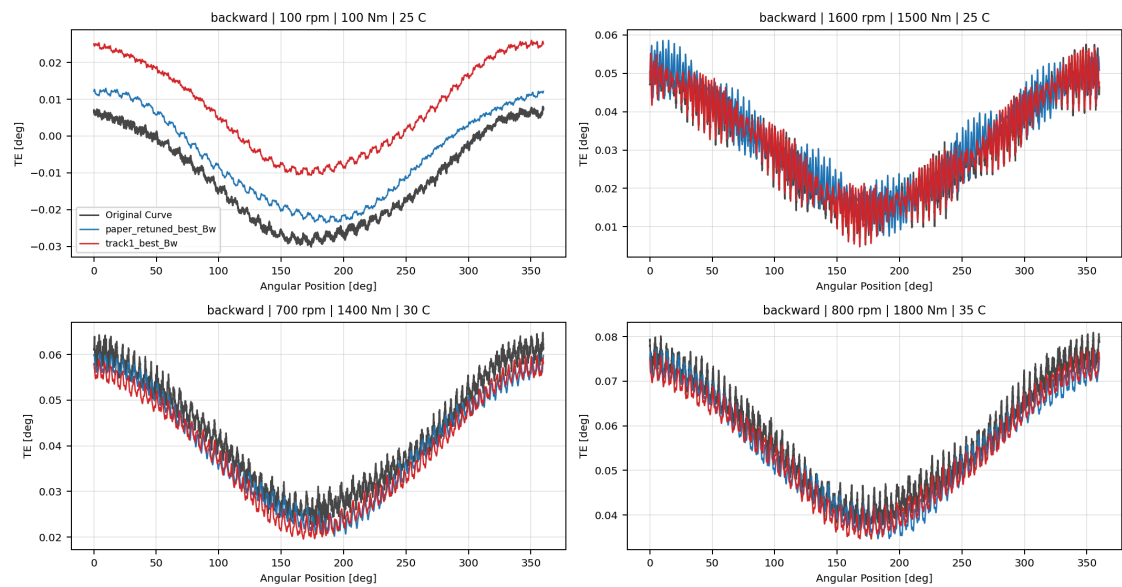


Forward Wave 1 Family Model Overlay Track 2 comparison

# Comparison Gallery - Backward Reference Model Overlay

Included models: `paper_retuned_best_Bw` , `track1_best_Bw` .

Backward Reference Model Overlay

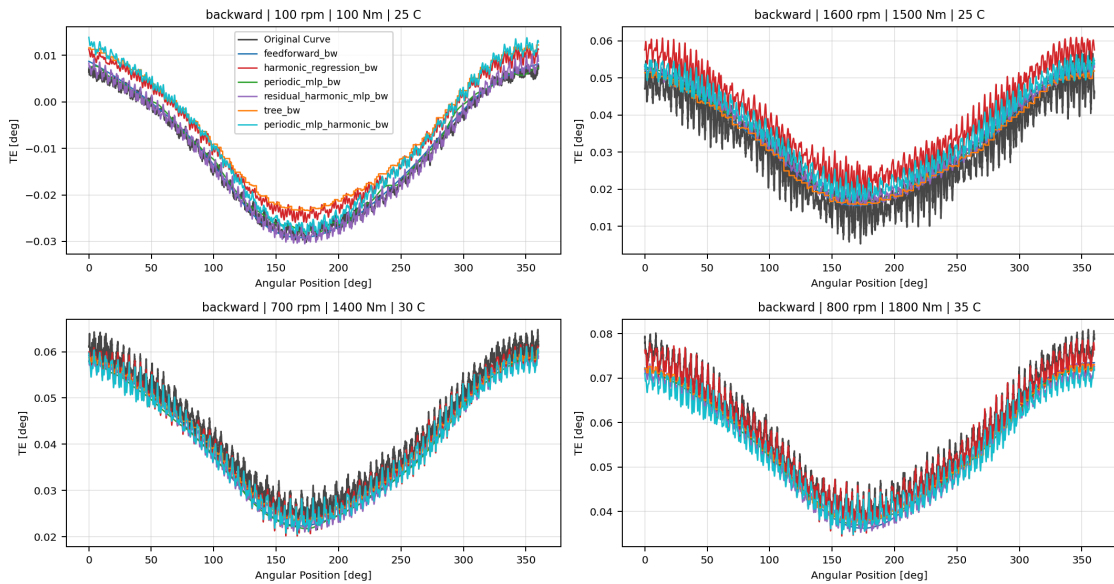


Backward Reference Model Overlay Track 2 comparison

## Comparison Gallery - Backward Wave 1 Family Model Overlay

Included models: `feedforward_bw` , `harmonic_regression_bw` , `periodic_mlp_bw` , `residual_harmonic_mlp_bw` , `tree_bw` , `periodic_mlp_harmonic_bw` .

Backward Wave 1 Family Model Overlay

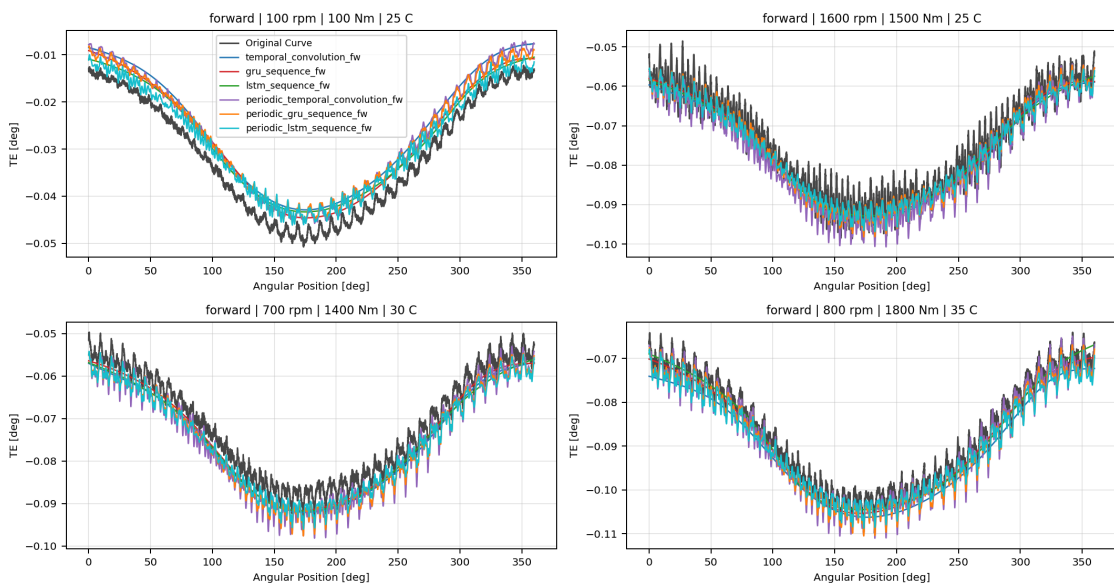


Backward Wave 1 Family Model Overlay Track 2 comparison

## Comparison Gallery - Forward Wave 2 Temporal Model Overlay

Included models: `temporal_convolution_fw` , `gru_sequence_fw` , `lstm_sequence_fw` , `periodic_temporal_convolution_fw` , `periodic_gru_sequence_fw` , `periodic_lstm_sequence_fw` .

Forward Wave 2 Temporal Model Overlay

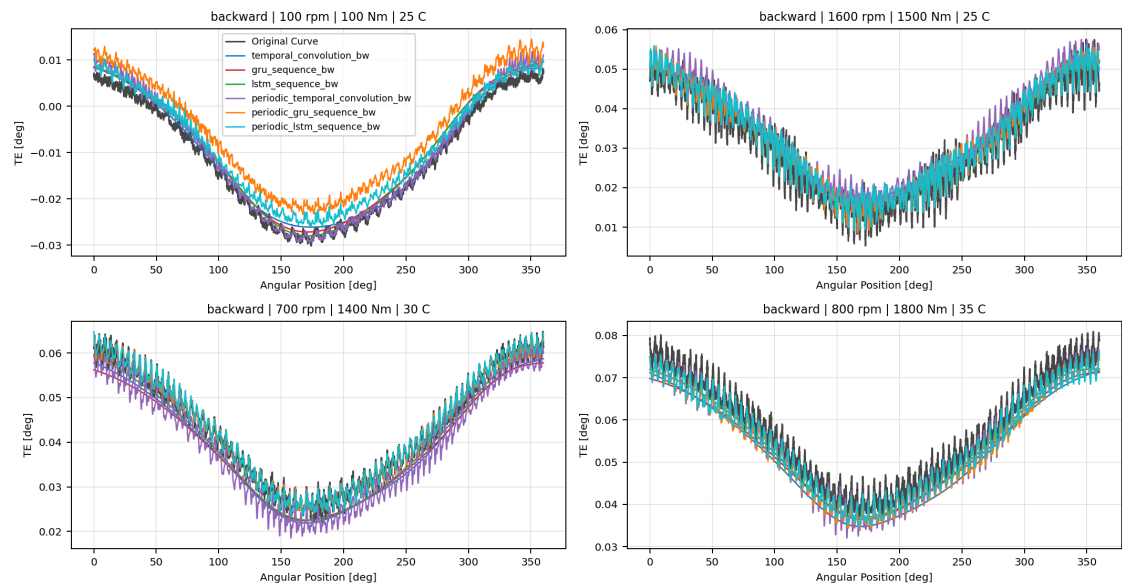


Forward Wave 2 Temporal Model Overlay Track 2 comparison

# Comparison Gallery - Backward Wave 2 Temporal Model Overlay

Included models: `temporal_convolution_bw` , `gru_sequence_bw` , `lstm_sequence_bw` , `periodic_temporal_convolution_bw` , `periodic_gru_sequence_bw` , `periodic_lstm_sequence_bw` .

Backward Wave 2 Temporal Model Overlay

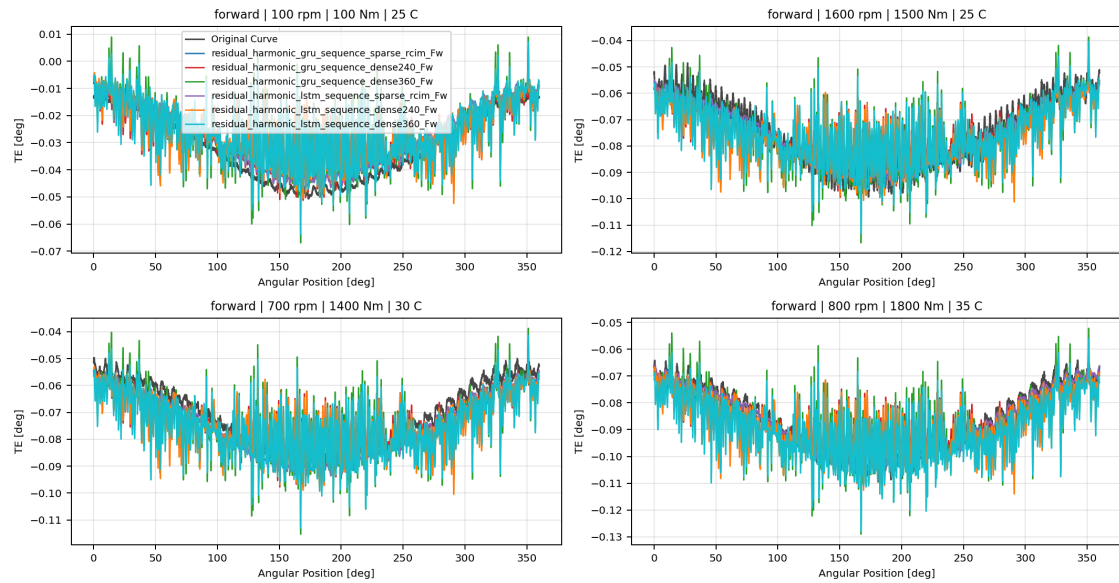


Backward Wave 2 Temporal Model Overlay Track 2 comparison

## Comparison Gallery - Forward Wave 2C Residual Harmonic Temporal Overlay

Included models: `residual_harmonic_gru_sequence_sparse_rcim_Fw` ,  
`residual_harmonic_gru_sequence_dense240_Fw` , `residual_harmonic_gru_sequence_dense360_Fw` ,  
`residual_harmonic_lstm_sequence_sparse_rcim_Fw` , `residual_harmonic_lstm_sequence_dense240_Fw` ,  
`residual_harmonic_lstm_sequence_dense360_Fw` .

Forward Wave 2C Residual Harmonic Temporal Overlay

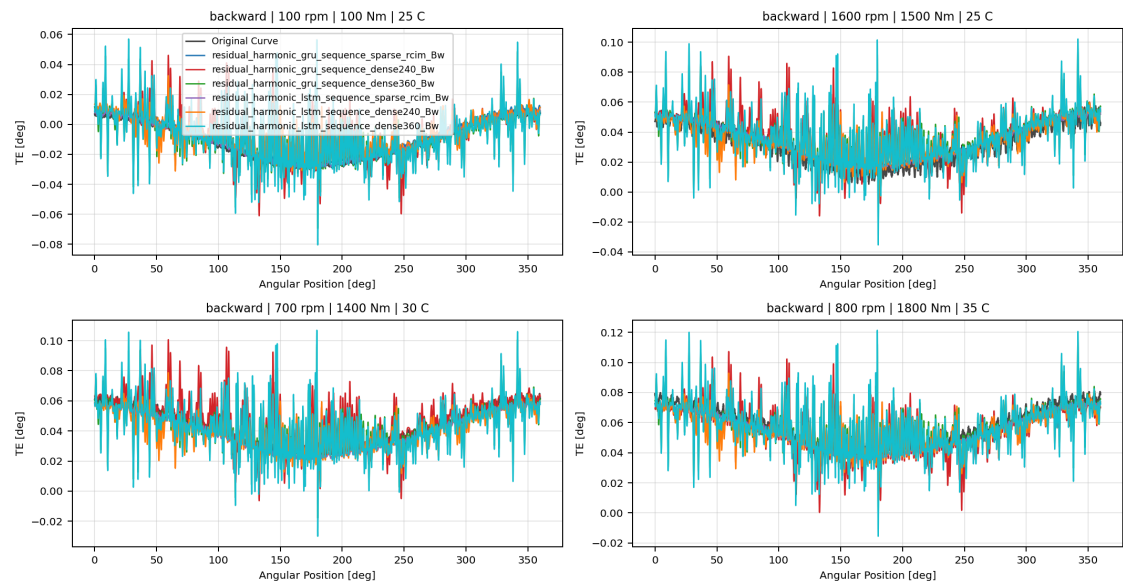


Forward Wave 2C Residual Harmonic Temporal Overlay Track 2 comparison

# Comparison Gallery - Backward Wave 2C Residual Harmonic Temporal Overlay

Included models: residual\_harmonic\_gru\_sequence\_sparse\_rcim\_Bw , residual\_harmonic\_gru\_sequence\_dense240\_Bw , residual\_harmonic\_gru\_sequence\_dense360\_Bw , residual\_harmonic\_lstm\_sequence\_sparse\_rcim\_Bw , residual\_harmonic\_lstm\_sequence\_dense240\_Bw , residual\_harmonic\_lstm\_sequence\_dense360\_Bw .

Backward Wave 2C Residual Harmonic Temporal Overlay



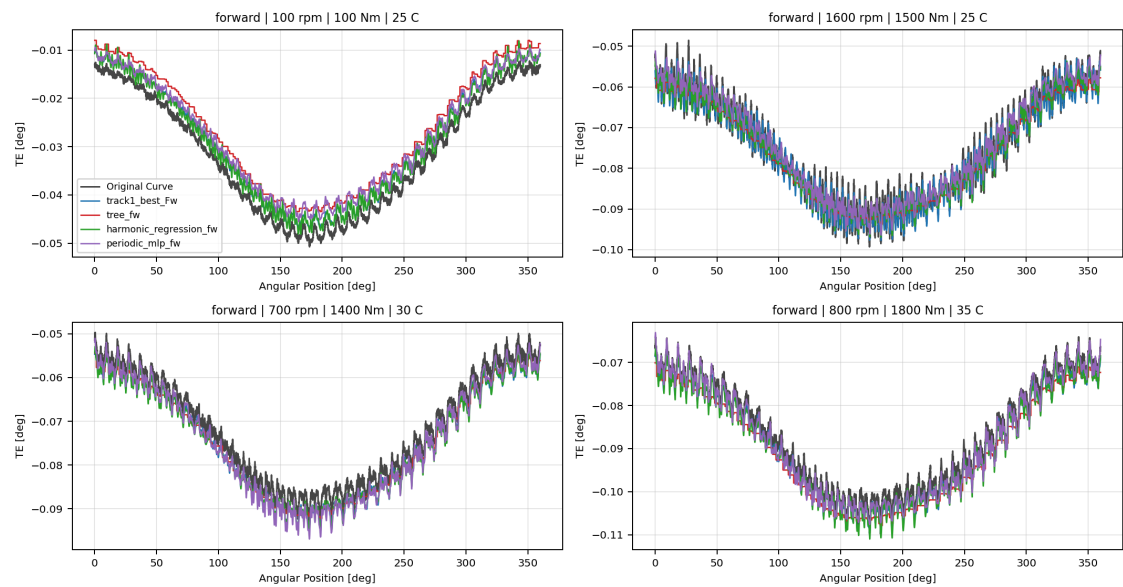
Backward Wave 2C Residual Harmonic Temporal Overlay Track 2 comparison



# Comparison Gallery - Forward Track 1 And Screened Wave 1 Overlay

Included models: track1\_best\_Fw , tree\_fw , harmonic\_regression\_fw , periodic\_mlp\_fw .

Forward Track 1 And Screened Wave 1 Overlay

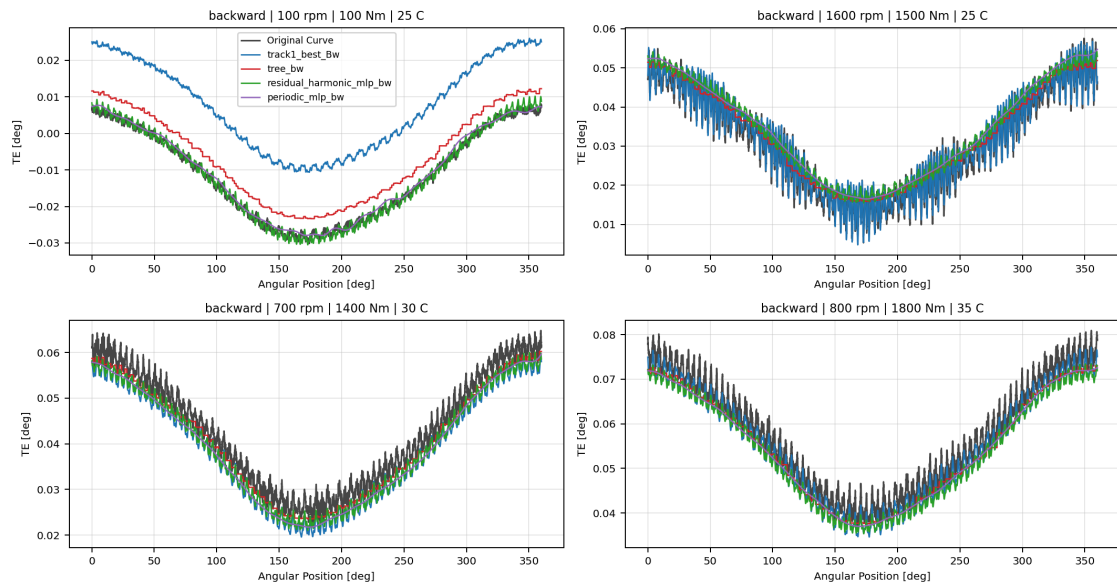


Forward Track 1 And Screened Wave 1 Overlay Track 2 comparison

## Comparison Gallery - Backward Track 1 And Screened Wave 1 Overlay

Included models: `track1_best_Bw`, `tree_bw`, `residual_harmonic_mlp_bw`, `periodic_mlp_bw`.

Backward Track 1 And Screened Wave 1 Overlay

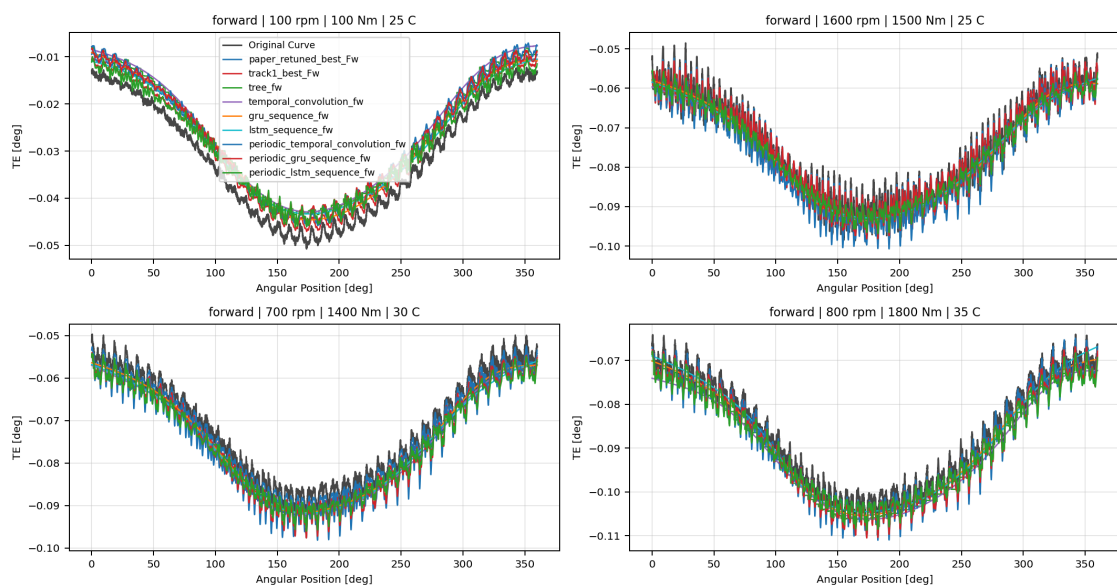


Backward Track 1 And Screened Wave 1 Overlay Track 2 comparison

## Comparison Gallery - Forward Reference Tree And Wave 2 Overlay

Included models: `paper_retuned_best_Fw`, `track1_best_Fw`, `tree_fw`, `temporal_convolution_fw`, `gru_sequence_fw`, `lstm_sequence_fw`, `periodic_temporal_convolution_fw`, `periodic_gru_sequence_fw`, `periodic_lstm_sequence_fw`.

Forward Reference Tree And Wave 2 Overlay

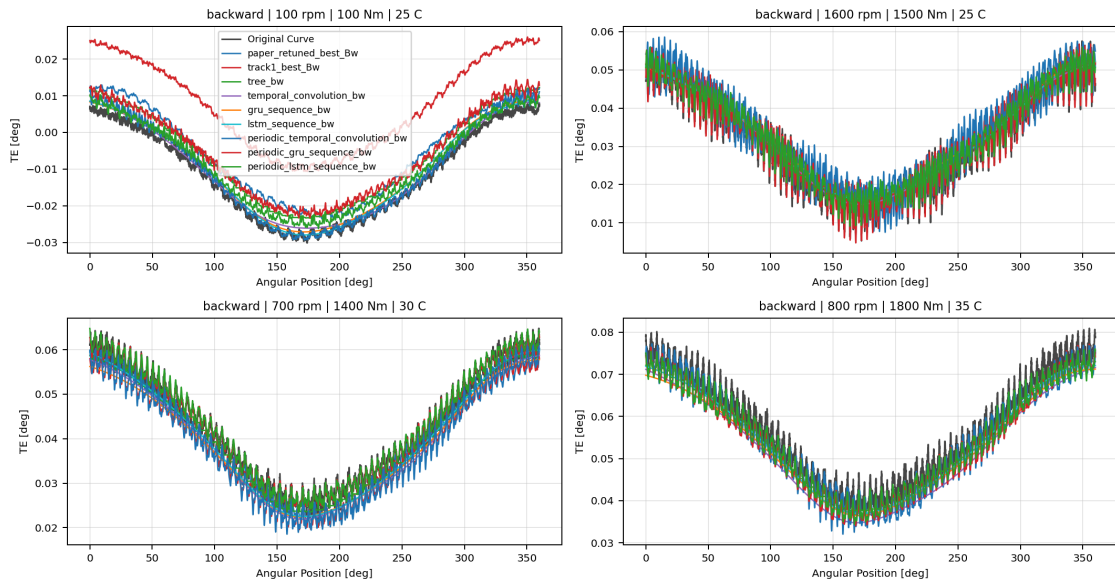


Forward Reference Tree And Wave 2 Overlay Track 2 comparison

## Comparison Gallery - Backward Reference Tree And Wave 2 Overlay

Included models: `paper_retuned_best_Bw`, `track1_best_Bw`, `tree_bw`, `temporal_convolution_bw`, `gru_sequence_bw`, `lstm_sequence_bw`, `periodic_temporal_convolution_bw`, `periodic_gru_sequence_bw`, `periodic_lstm_sequence_bw`.

Backward Reference Tree And Wave 2 Overlay

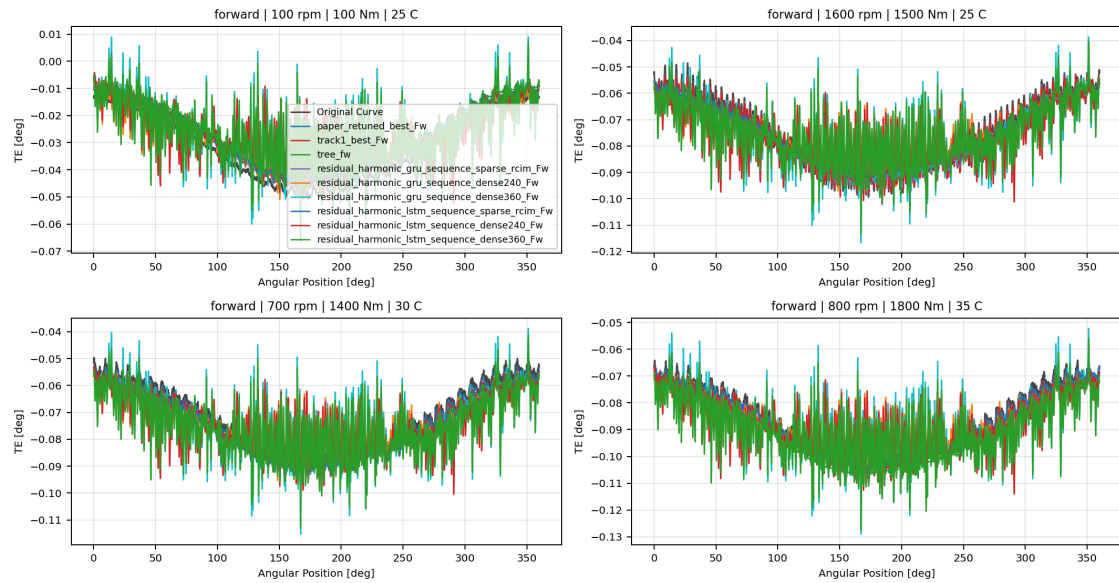


Backward Reference Tree And Wave 2 Overlay Track 2 comparison

## Comparison Gallery - Forward Reference Tree And Wave 2C Overlay

Included models: `paper_retuned_best_Fw`, `track1_best_Fw`, `tree_fw`,  
`residual_harmonic_gru_sequence_sparse_rcim_Fw`, `residual_harmonic_gru_sequence_dense240_Fw`,  
`residual_harmonic_gru_sequence_dense360_Fw`, `residual_harmonic_lstm_sequence_sparse_rcim_Fw`,  
`residual_harmonic_lstm_sequence_dense240_Fw`, `residual_harmonic_lstm_sequence_dense360_Fw`.

Forward Reference Tree And Wave 2C Overlay

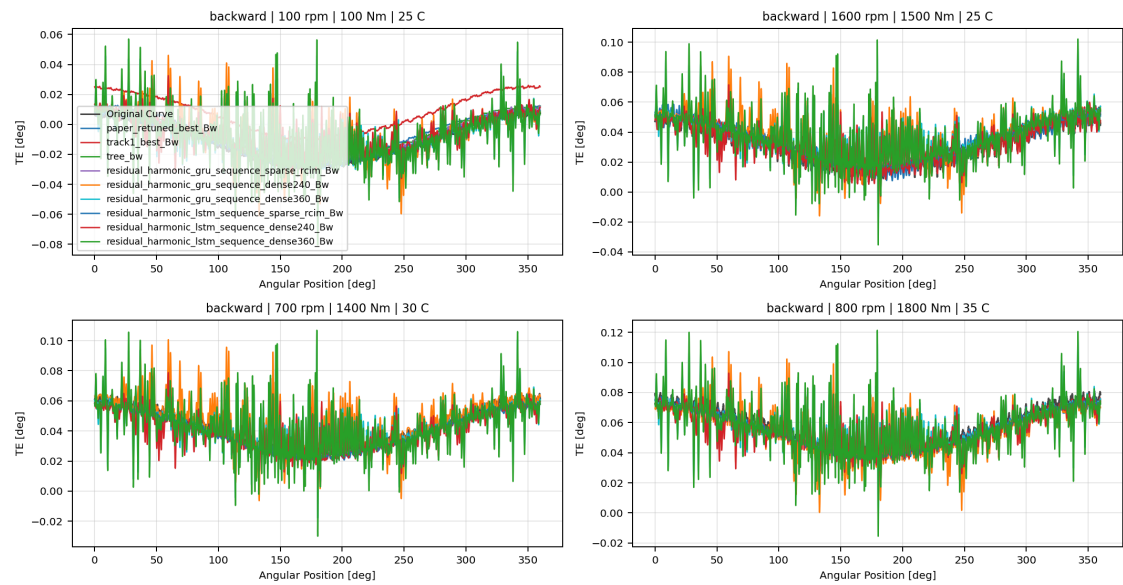


Forward Reference Tree And Wave 2C Overlay Track 2 comparison

# Comparison Gallery - Backward Reference Tree And Wave 2C Overlay

Included models: `paper_retuned_best_Bw` , `track1_best_Bw` , `tree_bw` ,  
`residual_harmonic_gru_sequence_sparse_rcim_Bw` , `residual_harmonic_gru_sequence_dense240_Bw` ,  
`residual_harmonic_gru_sequence_dense360_Bw` , `residual_harmonic_lstm_sequence_sparse_rcim_Bw` ,  
`residual_harmonic_lstm_sequence_dense240_Bw` , `residual_harmonic_lstm_sequence_dense360_Bw` .

Backward Reference Tree And Wave 2C Overlay



Backward Reference Tree And Wave 2C Overlay Track 2 comparison

## Output Artifacts

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- output directory: `output\validation_checks\track2_multi_model_curve_comparison_report\2026-06-04-17-37-56__track2_multi_model_curve_comparison_report` ;
- summary YAML: `output\validation_checks\track2_multi_model_curve_comparison_report\2026-06-04-17-37-56__track2_multi_model_curve_comparison_report\track2_multi_model_curve_comparison_summary.yaml` ;
- metrics CSV: `output\validation_checks\track2_multi_model_curve_comparison_report\2026-06-04-17-37-56__track2_multi_model_curve_comparison_report\track2_multi_model_curve_comparison_metrics.csv` ;
- report Markdown: `doc\reports\analysis\track2\multi_model_curve_comparison_report\[2026-06-04]\track2_multi_model_curve_comparison_report.md` .